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ALICE L WALTON SCHOOL OF MEDICINE

Local Control of the Microcirculation

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Adapted from: Local Control of the Microcirculation

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Describe the myogenic mechanism involved in the autoregulation of blood flow.
2. Describe the metabolic control of blood flow by factors PO_2, PCO_2, pH, lactate, adenosine, and $[K^+]_{plasma}$.
3. Describe the paracrine control of blood flow.
4. Describe the sympathetic and hormonal control of blood flow.
5. Explain the mechanism by which local blood vessels release nitric oxide.

CASE CONNECTION

You and your friend decide that you are going to start running to help manage stress and take off a few extra pounds. As you both reach mile 4, you start thinking about how our our leg muscles can perform for such a long time.

What role do local tissue factors play in the microcirculation to increase blood flow to areas in need, such as your leg muscles as you run? Consider your answer as you read, and we'll revisit the question at the end of the brick.

GO TO CONCLUSION ↓

Introduction

The body uses precise control mechanisms to regulate blood flow, ensuring tissues receive adequate oxygen and nutrients while maintaining systemic vascular resistance and arterial pressure. This **local regulation of blood flow** occurs when the arteriolar smooth muscle cells are either stretched or respond to environmental stimuli; they either vasoconstrict to reduce blood flow or vasodilate to increase blood flow in the area.

So we are talking abo[illegible]nd this is based on the flow equation, where Flow = Pressure gradient over Resistance (Q = ΔP/R). Central to this are the marked flow changes with changes in radius and the resulting resistance. A higher resistance, or a smaller radius, means less blood flow.

Previously, we applied this flow concept to the entire circulatory system; in this Brick, the focus is on the flow to different organs or tissues. There are 2 broad stimuli for the **local control of blood flow**:

- **Intrinsic regulation:** is the control of LOCAL blood flow to organs/tissues, based on either vessel stretch or local control factors.
- **Extrinsic regulation** is mainly concerned with controlling mean arterial pressure (MAP) and total peripheral resistance. Basically, the blood pressure has to be high enough to allow for tissue/organ perfusion, and directing flows to different areas.

These processes vary in dominance depending on the vascular bed—skeletal muscle, brain, skin, or lungs. This brick explores each control mechanism in depth and highlights their clinical significance.

Instrinsic— Myogenic control of local blood flow (LO1)

Myogenic control is an intrinsic control mechanism in the blood vessels activated in response to blood pressure (Figure 1A). This control is a sustained muscle contraction (tone) generated by the **vascular smooth muscle** itself (myogenic) without the influence of external stimuli like nerves or hormones. It works by regulating the degree of vascular smooth muscle tone (degree of contraction) in response to the stretch of the blood vessel. Changes in muscle tone regulate the vas[illegible]ood flow to the tissue.

If blood pressure increases, the vascular smooth muscle cells stretch; this stretch causes the cells to reflexively contract, narrowing the vessel (Figure 1B). The local vascular resistance increases, reducing flow, and prevents an increase in blood flow to maintain a constant blood flow to that area (**autoregulation**).

The reverse happens in moderate hypotension or with low blood pressure. If the MAP decreases, there is reduced stretch, leading to relaxation (dilation), and the lowered resistance maintains flow in the normal range. Note that this autoregulatory response only works within the blood pressure range of 60-150 mmHg.

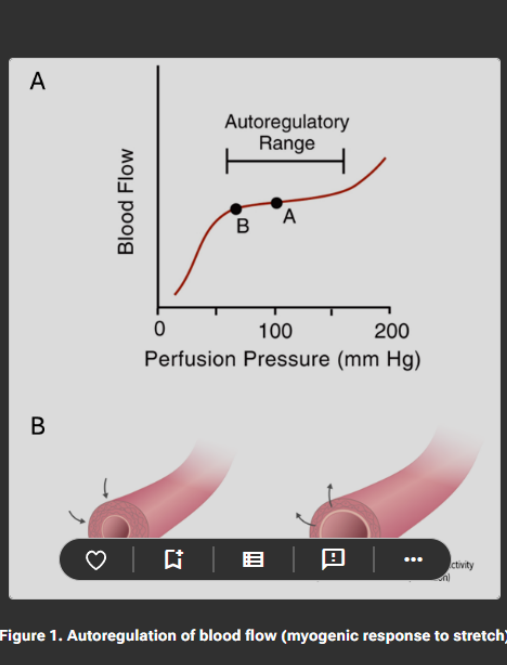


Figure 1. Autoregulation of blood flow (myogenic response to stretch)

What is the mechanism? Increased mean arterial pressure (MAP) stretches the blood vessel walls, activating stretch-sensitive Ca^{2+} ion channels, and the influx of calcium into the cells causes smooth muscle contraction. When the signal abates, the excess calcium is transported to the outside of the cell, or pumped back into intracellular stores (sarcoplasmic reticulum). This vasoconstriction of the arterioles increases resistance and normalizes blood flow.

CLINICAL REASONING

Cerebral autoregulation keeps blood flow stable between MAP 60–160 mmHg. Below this, perfusion fails, causing ischemia. Above this, intracranial pressure increases. **Shock states (e.g., hemorrhagic):** Myogenic control can fail under sustained hypotension, contributing to multi-organ dysfunction.

How does a rise in arterial blood pressure affect the blood vessel wall?

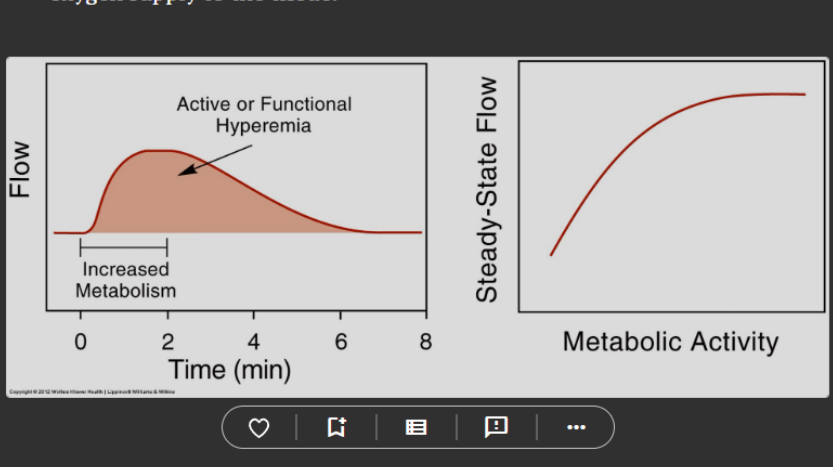
What is the myogenic response?

Intrinsic—Metabolic Control of Local Blood Flow (LO2)

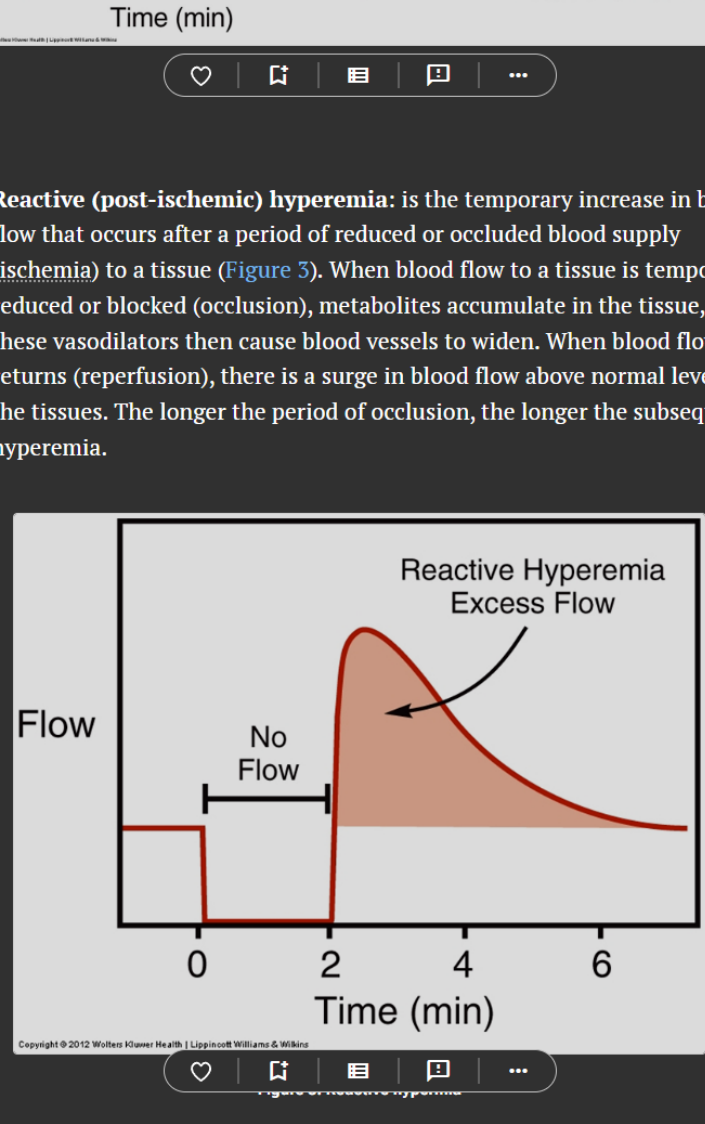
Tissues can control their local blood flow. Circulating molecules have the greatest influence on the control of blood flow in the microcirculation. When a tissue becomes more metabolically active, it clearly needs more blood flow and nutrients. To accomp[illegible]ances (metabolites, endothelial secretions, or paracrine factors) that result in local increases in blood flow. This increased local blood flow (hyperemia) delivers more oxygen and nutrients to match the tissue's higher metabolic demands.

Active and Reactive Hyperemia are the terms for two different types of blood flow in response to metabolic vasodilators:

- **Active, metabolic or functional hyperemia** (Figure 2): is where increased metabolic activity (e.g, exercise) results in local accumulation of metabolite concentrations, which in turn increase local blood flow to the region. This explains the increase in blood flow to skeletal and cardiac muscle during exercise, fight or flight, and to the brain during cognitive tasks. Examples of these vasodilator metabolites are CO_2, H^+, adenosine, and K^+, and they are able to override sympathetic activity (functional sympatholysis- more later). Lastly, and importantly, low PO_2 levels or hypoxia, results in vasodilation to increase blood flow to help restore oxygen supply to the tissue.



- **Reactive (post-ischemic) hyperemia**: is the temporary increase in blood flow that occurs after a period of reduced or occluded blood supply (ischemia) to a tissue (Figure 3). When blood flow to a tissue is temporarily reduced or blocked (occlusion), metabolites accumulate in the tissue, and these vasodilators then cause blood vessels to widen. When blood flow returns (reperfusion), there is a surge in blood flow above normal levels to the tissues. The longer the period of occlusion, the longer the subsequent hyperemia.



What mediators contribute to active hyperemia?

Metabolites involved in local control of blood flow

Let's take a closer look at the metabolites involved in local control of blood flow (Figure 4):

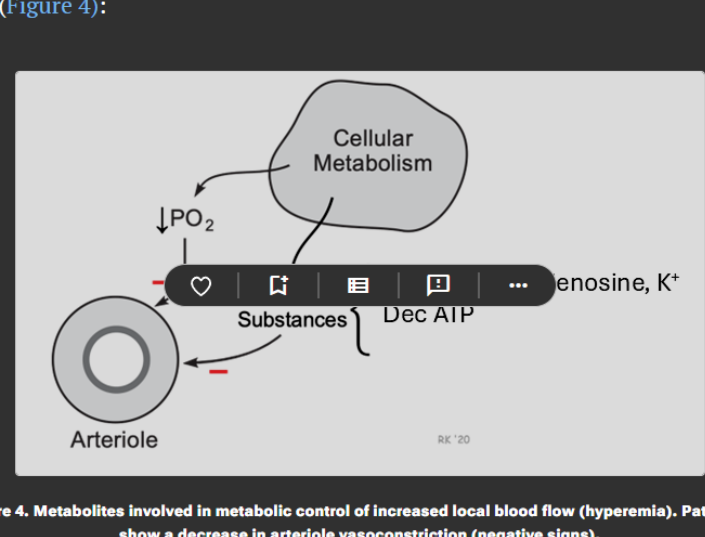


Figure 4. Metabolites involved in metabolic control of increased local blood flow (hyperemia). Pathways show a decrease in arteriole vasoconstriction (negative signs).

- **Low PO_2:** Low partial pressures of oxygen (a measure of dissolved O_2 in the blood) influences blood flow. The amount of O_2 in the blood is typically measured as arterial levels (PaO_2), which is normally ~100 mm Hg. Low PaO_2 levels in the blood < 60 mm Hg are hypoxic to tissues. Hypoxia results in vasodilation to increase blood flow to help restore oxygen supply to the tissue. The exception is the lungs, where hypoxia constricts pulmonary vessels.
- **High PCO_2:** levels are a potent vasodilator, especially in the brain. Cerebral blood vessels are especially sensitive to changes in PCO_2 and H^+, making arterial PCO_2 a key regulator of brain blood flow.
- **Increased H^+:** reflects vasodilation due to lowered pH. This acidosis-mediated vasodilation is due to increased metabolism (generating more CO_2) or anaerobic metabolism (eg, lactic acidosis). It is important to note that blood H^+ does not directly affect cerebral blood flow, as it cannot cross the blood-brain barrier (BBB).
- **Adenosine and ATP:** accumulates with an increase in the breakdown of ATP, particularly in skeletal and cardiac muscles. It strongly correlates with metabolic activit[illegible]n the kidneys however, adenosine functions as a vasoconstrictor.
- **Potassium (K^+):** accumulates extracellularly when ATP levels drop since the Na^+/K^+-ATPase pump activity requires ATP to transport potassium into cells. With increased skeletal muscle and brain activity, extracellular potassium rises from approximately 4 mM to 9 mM. Unexpectedly, this elevated potassium causes vascular smooth muscle to relax rather than contract, resulting in vasodilation and increased blood flow to the active tissues.

CLINICAL REASONING

Why is blood flow to the heart, brain, and skeletal muscle not decreased during sympathetic vasoconstriction, which is expected during "fight or flight"?.

Functional sympatholysis is where local metabolic factors, such as nitric oxide, override sympathetic-induced vasoconstriction. Initial α1-receptor sympathetic stimulation causes transient vasoconstriction, but this is quickly reversed by locally released metabolites, causing vasodilation. This overriding mechanism is essential in skeletal muscle, heart, and brain, ensuring these critical tissues maintain adequate perfusion during high sympathetic activity states like "fight or flight" responses (e.g., exercise).

Paradoxical vasodilation by K^+

We would expect that high extracellular K^+ (hyperkalemia) would typically reduce K^+ efflux and [illegible]epolarization does not happen in vascular smooth muscle, and paradoxically hyperkalemia actually causes vasodilation. This is because the increased K^+ levels open specific K^+ ion channels, allowing more K^+ to exit the vascular smooth muscle. This hyperpolarizes the membrane potential near the potassium equilibrium (E_K), prevents significant depolarization, and leads to closure of L-type Ca^{2+} channels. As a result, less extracellular calcium enters the cell, and the smooth muscle relaxes. Hyperkalemia also boosts Na^+/K^+ pump activity, increasing K^+ uptake into the cell, helping to maintain this repolarization effect.

Which of the local tissue factors would be increased in concentration in the leg muscles of someone who just finished running a mile?

Paracrine Control of Blood Flow (LO3)

Let's focus on paracrine signaling molecules that precisely control blood flow by adjusting vascular[illegible]creted, where they diffuse to have local effects on neighboring vascular smooth muscle cells and trigger either relaxation (vasodilation) or contraction (vasoconstriction) (Figure 5). As such, they exert local control on blood flow.

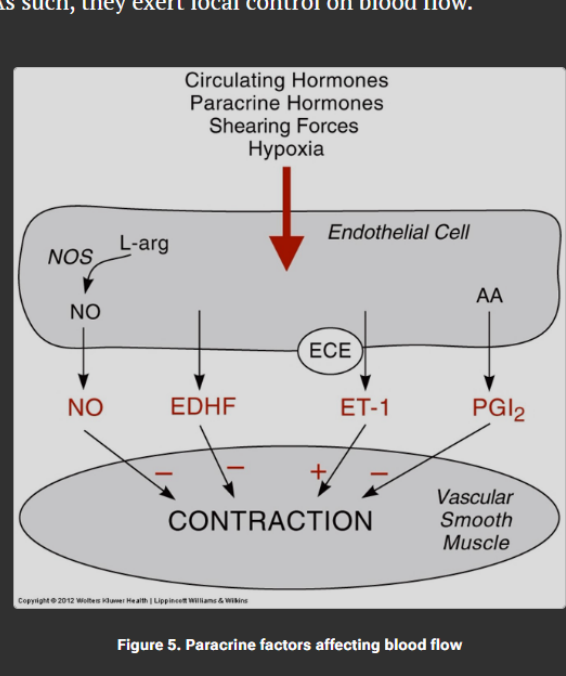


Figure 5. Paracrine factors affecting blood flow

Nitric Oxide (NO) (LO5)

Nitric oxide (NO) is a paracrine regulator of blood flow involved in vasodilation. NO release can be locally stimulated by **shear stress**, which is the force applied to the v[illegible]flowing parallel to the vessel wall (Figure 6). This activates endothelial nitric oxide synthase (eNOS) and increases NO synthesis. Other agents can also increase NOS and NO, including adenosine and bradykinin.

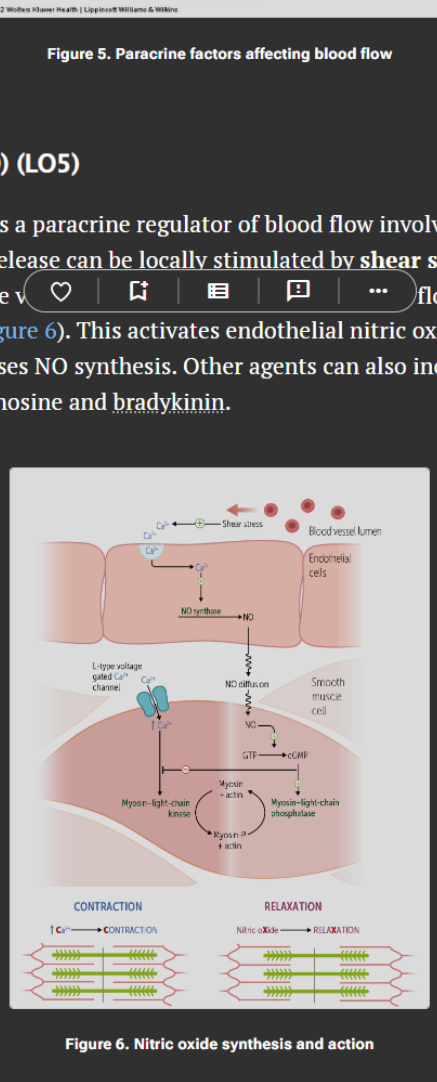


Figure 6. Nitric oxide synthesis and action

The steps in the pathway with a paracrine effect, are:

1. Nitric oxide synthase within the endothelial cells (eNOS) uses L-arginine and other cofactors to make nitric oxide.
2. NO diffuses local[illegible]
3. NO activates guanylyl cyclase to turn guanosine triphosphate (GTP) into cyclic guanosine monophosphate (cGMP).
4. cGMP promotes smooth muscle relaxation by enhancing calcium reuptake into the sarcoplasmic reticulum (SR). This activates potassium channels (causing hyperpolarization) and myosin–light-chain phosphatase (MLCP).
5. MLCP dephosphorylates myosin light chains, leading to smooth muscle relaxation.

CLINICAL CORRELATION

Angina and NO: Nitroglycerin is an NO donor used to treat chest pain in angina pectoris (it is also used as an explosive). Nitroglycerin was initially used in the mass manufacturing of dynamite, where factory workers often developed headaches during the week that vanished on weekends, while those with heart disease noticed relief from chest pain at work but not on weekends. Physicians later realized these effects were due to nitroglycerin's vasodilatory action, leading to its medical use to widen blood vessels, improve circulation, and protect the heart from damage and cell death.

Angina is caused by inadequate blood flow to the heart myocardium. Nitroglycerin generates nitric oxide. The NO increases cGMP in vascular smooth muscle, preferentially causing vasodilation over arteriodilation. Venodilation (dilation of veins) reduces cardiac workload through decreased venous return and preload (blood pooling in capacitance veins). Arterial dilation simultaneously reduces afterload (lowering peripheral resistance) and improves coronary perfusion. This dual mechanism decreases myocardial oxygen demand while increasing oxygen supply, effectively relieving chest pain.

NO made in the vascular endothelium is most important in regulating local blood flow. However, circulating NO has a role in the systemic control of blood pressure, continually modulating basal vascular tone, and is involved in exercise, penile erection, and inflammation.

CLINICAL CORRELATION

Nitric Oxide in Inflammation and Endotoxin Shock: Inflammation and endotoxin shock involve activation of inducible nitric oxide synthase (iNOS), which produces more NO than endothelial NOS (eNOS). During **inflammatory responses**, mediators like bradykinin, substance P, and thrombin activate both eNOS and iNOS, increasing NO production. **Bacterial endotoxins** stimulate monocytes and macrophages to release interferon-gamma, which potently activates iNOS. The resulting excessive NO production can lead to severe hypotension, manifesting as endotoxin shock

Other Vasodilators (LO3)

- **Prostacyclin (PGI_2)**: is a prostaglandin produced in response to thrombin that dilates vessels in that specific region. It also inhibits platelet aggregation, preventing uncontrolled thrombosis.
- **Histamine**: is released from mast cells during allergic inflammation. It dilates arterioles and veins and increases venular permeability, causing redness and edema at the local site. This may facilitate immune cell migration, and dilution of toxins and irritants.
- **Bradykinin:** cau[illegible], creating redness and warmth while increasing vascular permeability that leads to swelling. It also sensitizes nerve endings, producing pain at inflamed sites.

What is the effect of nitric oxide release from endothelial cells?

Vasoconstrictors (LO3)

- **Endothelin-1**: contributes to basal vascular tone and is elevated in pathological states like hypoxia, preeclampsia, and cardiac failure.
- **Thromboxane A_2:** induces vasoconstriction of vascular smooth muscle. It serves as a positive-feedback mediator during platelet activation.
- **Serotonin:** is released from platelets during clotting to cause vasoconstriction and stop bleeding.

What role does endothelin-1 play in vascular tone?

CLINICAL CORRELATION

Paracrine factors in disease states.

Hypoxia contributes to pulmonary hypertension at high altitudes.

Preeclampsia involves elevated endothelin levels leading to hypertension.

Anaphylaxis involves histamine release causing systemic vasodilation and capillary leakage, resulting in hypotension and edema.

Thrombosis (blood clot formation within vessels) - thromboxane promotes platelet aggregation. Low-dose aspirin prevents thrombosis by inhibiting thromboxane formation in platelets. While atherosclerosis itself doesn't directly form a thrombus, the rupture of an atheromatous plaque triggers thrombosis through platelet activation and clot formation.

Extrinsic — Systemic Nervous System and Hormonal Control of Microcirculation (LO4)

Autonomic control

Sympathetic nervous system activity primarily causes vasoconstriction via alpha-1 receptors, especially in the skin and gastrointestinal tract. Vasodilation in skeletal muscle is mediated by beta-2 receptors, predominantly in response to circulating epinephrine rather than direct sympathetic nerve activity. Baseline sympathetic tone is essential for maintaining normal systemic vascular resistance.

Parasympathetic nervous system does not have major effects on local control of blood flow (microcirculation). Its effects on blood vessels have a limited distribution, primarily affecting salivary glands and genitals. The mechanism involves acetylcholine binding to muscarinic receptors, which increases nitric oxide production and subsequently causes vasodilation in these specific tissues.

How do $α_1$ and $β_2$ receptors differ in their vascular effects?

Hormonal regulation

Epinephrine (Epi) is released from the adrenal medulla during the fight-or-flight response and binds to both $β_1$ and $β_2$ adrenergic receptors on target organs. In skeletal muscle vasculature, epinephrine preferentially activates $β_2$ receptors, causing vasodilation and a decrease in systemic vascular resistance (total peripheral resistance). Simultaneously, $β_1$ receptor activation in the heart increases heart rate and contractility, leading to increased cardiac output and systolic pressure (Figure 7). These opposing effects often result in little or no increase in mean arterial pressure (MAP), and not triggering a baroreceptor response. At high (pharmacologic) concentrations (not shown), epinephrine increasingly activates $α_1$, shifting the effects to net vasoconstriction and increased SVR. *(Shortcut: Epi = β at low dose, α at high dose).*

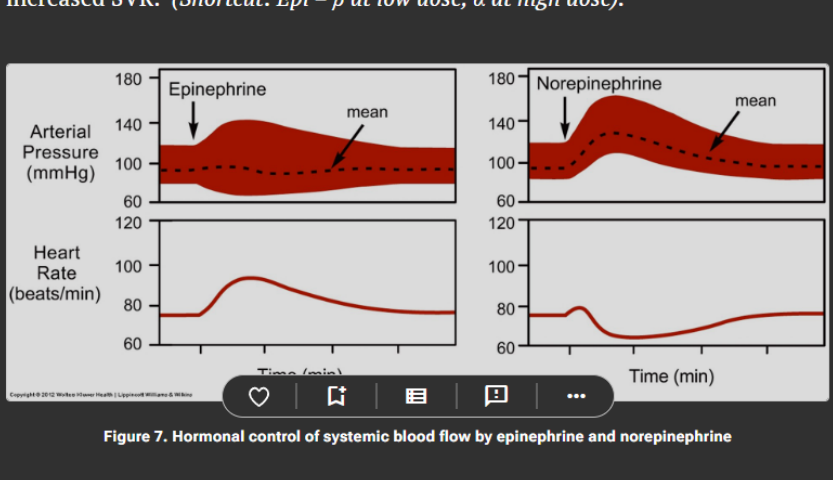


Figure 7. Hormonal control of systemic blood flow by epinephrine and norepinephrine

Norepinephrine (NE) is released primarily from sympathetic postganglionic neurons. NE binds adrenergic receptors with a strong preference for $α_1$ over β receptors (α ≫ β). Activation of $α_1$ receptors causes widespread vasoconstriction, increasing both systolic and diastolic pressures. NE also activates β1 receptors in the heart, producing a transient increase in heart rate and contractility (seen as a brief HR "blip" in Figure 7). However, the resulting rise in arterial pressure activates the baroreceptor reflex, increasing vagal tone and causing reflex bradycardia. Thus, the net effect of NE is vasoconstriction with decreased heart rate. *(Shortcut: NE = $α_1$ vasoconstriction + reflex ↓HR.)*

Angiotensin II, released through the renin-angiotensin-aldosterone system (RAAS), and **vasopressin (ADH)**, released through osmolarity regulation, are important vasoconstrictors that increase total peripheral resistance and mean arterial pressure throughout the body.

Extrinsic regulation—In summary, extrinsic regulation is mainly concerned with controlling mean arterial pressure (MAP) and total peripheral resistance. It maintains blood pressure in ALL parallel arteries for tissue/organ perfusion, and in this way, it also affects local blood flow to tissues (but on a grander scale).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to the role that local tissue factors play in the microcirculation, how do they increase blo[illegible]

You go through the sequence of events in your mind in three steps. In exercise, muscles need more blood flow and oxygen delivery. The underlying physiologic response to these needs is vasodilation. The local tissue factors generated during exercise that increase dilation in the microcirculation include low regional oxygen levels (due to high levels of oxygen extracted by the working muscle), increased CO_2 (due to an increase in CO_2 generated by the working muscle), decreased pH (from lactic acid accumulation), and increased adenosine and extracellular potassium. Phew!

Summary

- Local regulation of blood flow involves intrinsic and extrinsic mechanisms to ensure tissues receive adequate oxygen and nutrients.
- **Intrinsic** control of blood flow refers to the regulation that occurs locally within tissues and is independent of systemic neural or hormonal input.
- Autoregulation (intrinsic) is where the body maintains steady blood flow to critical tissues over a range of normal systemic blood pressures, via the myogenic response/ reflex. This response does not work outside of the normal, physiological blood pressure range (~60 - 150 mmHg).
- Myogenic control involves vascular smooth muscle contraction in response to blood pressure changes, maintaining consistent blood flow within a specific pressure range.
- Active hyperemia is the shunting of blood and an increase in blood flow to specific regions in response to elevated metabolic activity. As tissues work harder—during exercise or cognitive tasks—they release vasodilator metabolites that[illegible]g oxygen and nutrient delivery. This response ensures that tissue perfusion matches metabolic rate.
- Reactive hyperemia refers to the transient surge in blood flow following a brief period of occlusion. Downstream of the occlusion, accumulated metabolites trigger local vasodilation (tissues are starving). When the occlusion is released, there is a temporary but pronounced increase in flow.
- Metabolic control relies on the accumulation of vasodilator metabolites like CO_2, H^+, adenosine, lactate and extracellular K^+, and decreased O_2 levels (hypoxia at tissues), to increase blood flow during heightened metabolic activity.
- Paracrine factors such as nitric oxide, prostacyclin, and bradykinin modulate vascular tone, with nitric oxide (released in response to shear stress at endothelial cells) playing a key role in vasodilation.
- Extrinsic regulation, including sympathetic nervous system activity and hormones like epinephrine and angiotensin II, affects systemic blood flow and pressure.
- Sympathetic stimulation typically causes vasoconstriction via $α_1$ receptors, e.g. in the skin and gut, while $β_2$ receptor activation leads to vasodilation in skeletal muscle.
- Local control of blood flow can override sympathetic tone to increase perfusion to active tissues, a process known as **functional sympatholysis.**
- Epinephrine, norepinephrine, angiotensin II, and ADH (vasopressin) are key extrinsic mediators affecting the local blood flow to ALL organs and tissues.

Review Questions

1. Which of the follow[illegible]f nitric oxide to cause vasodilation?

A. ○ Guanylyl cyclase

B. ○ Myosin–light-chain phosphatase

C. ○ Oxygen

D. ○ Relaxation of the smooth muscle

E. ● Shear stress

Hide Explanation

The correct answer is E). Nitric oxide synthase can be stimulated by shear stress on the vascular endothelium or by inflammation, infection, adenosine, bradykinin, or other vasodilating factors. Incorrect answers, A). Nitric oxide activates guanylyl cyclase, not the other way around. B). Myosin–light-chain phosphatase is activated downstream of nitric oxide production. C). Oxygen does not play a role in NO synthesis. D). Relaxation of the smooth vessel is an effect, rather than a cause, of nitric oxide.

2. A **decrease** in which of the following will cause vasodilation?

A. ○ Adenosine

B. ○ Carbon d[illegible]

C. ○ Extracellular potassium

D. ○ Hydrogen ions

E. ● Oxygen

Hide Explanation

The correct answer is E). Low levels of oxygen reflect increased metabolism and result in vasodilation. Incorrect answers. A)., B) . C)., D)., and E). are all local tissue factors related to metabolic activity within the tissue, and an increase in these factors stimulates vasodilation of the microcirculation.

3. Which of the following vessels are primarily responsible for regulating blood flow at the tissue level?

A. ○ Aorta

B. ○ Large veins

C. ● Arterioles

D. ○ Capillaries

E. ○ Venules

Hide Explanation

The correct answer is C). These are resistance vessels with smooth muscle that regulate blood flow into capillaries by changing diameter. Incorrect answers A). Aorta is a conduit vessel, not involved in local flow regulation. B). Large veins are reservoir vessels affecting blood volume, and don't significantly influence resistance. D). Capillaries are the main site of exchange, not regulation. E). Venules are involved in drainage, not primary regulators.

4. Autoregulation of blood flow maintains a constant perfusion despite changes in which of the following:

A. ○ Heart rate

B. ○ Arterial oxygen content

C. ● Mean arterial pressure

D. ○ Capillary permeability

Hide Explanation

The correct answer is: C). Autoregulation keeps flow steady between ~60–160 mmHg by adjusting arteriolar tone. Incorrect answers: A). Heart rate influences cardiac output, not local flow regulation. B). O_2 content affects oxygen delivery, not flow autoregulation. D). Permeability affects exchange, not pressure regulation.

5. Which of the following paracrine factors is most associated with vasoconstriction rather than vasodilation?

A. ○ Nitric oxide (NO)

B. ○ Prostacyclin

C. ● Thromboxane A_2

D. ○ Histamine

E. ○ Bradykinin

Hide Explanation

The correct answer is C). Thromboxane A_2 is a paracrine factor that promotes vasoconstriction and platelet aggregation, typically released during platelet activation. Incorrect answers: A). Nitric oxide causes smooth muscle relaxation and vasodilation. B). Prostacyclin is a vasodilator that also inhibits platelet aggregation. D). Histamine is a vasodilator released during allergic reactions. E). Bradykinin is a vasodilator involved in inflammation and increased vascular permeability.

6. Which of the following best describes the vascular response to low-dose epinephrine?

A. ○ Widespre[illegible]ulation

B. ○ Decreased heart rate via beta-1 receptor stimulation

C. ● Vasodilation in skeletal muscle via beta-2 receptor stimulation

D. ○ Vasoconstriction in skeletal muscle via beta-2 receptor stimulation

Hide Explanation

The correct answer is C). Vasodilation in skeletal muscle via beta-2 receptor stimulation. Epinephrine has concentration-dependent effects on blood vessels. At physiological concentrations, epinephrine preferentially stimulates beta-2 receptors in skeletal muscle, causing vasodilation to enhance blood flow during sympathetic activation. At higher pharmacological concentrations, epinephrine preferentially activates alpha-1 receptors, resulting in vasoconstriction. Incorrect answers: A). Alpha-1 effects dominate at higher (pharmacologic) doses.
B). Beta-1 increases heart rate, not decreases it. D). Beta-2 stimulation causes vasodilation, not constriction.

How would you rate this Brick?

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Review Learning Objectives

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the myogenic mechanism involved in the autoregulation of blood flow.~~

02 ☑ ~~Describe the metabolic control of blood flow by factors PO_2, PCO_2, pH, lactate, adenosine, and $[K^+]_{plasma}$.~~

03 ☑ ~~Describe the paracrine control of blood flow.~~

04 ☑ ~~Describe the sympathetic and hormonal control of blood flow.~~

05 ☑ ~~Explain the mechanism by which local blood vessels release nitric oxide.~~

Objective Confidence: 5/5